# SIMATS ENGINEERING

## Lab Programs

**COURSE CODE: ECA1009** **COURSE NAME: Microprocessors & Microcontrollers**

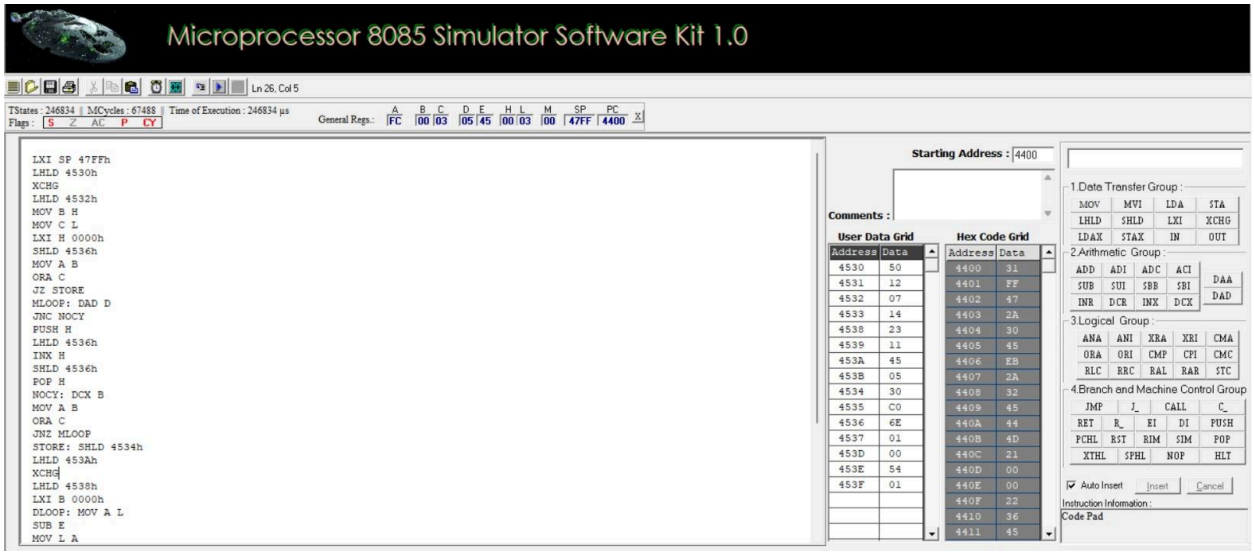
***Program D* – Implementation and Testing of 16-Bit Multiplication and Division Using Intel 8-Bit Microprocessor Assembly Language Programming**

**Aim:** To implement and test 16-bit multiplication and division using Intel 8085 assembly language programming.

**Algorithm:**

1. Initialize the stack pointer and load the 16-bit operands from memory.
2. Store the operands in the appropriate register pairs.
3. Perform 16-bit multiplication using repeated addition.
4. Store the multiplication result in the specified memory locations.
5. Perform 16-bit division using repeated subtraction and maintain the quotient and remainder.
6. Store the quotient and remainder in memory and halt the program.

**Execute:**




**Result:** The 16-bit multiplication and division program was executed using the 8085 simulator, and the results were stored in the specified memory locations.